

E1 LightGBM — Phase 1.5B Operational Metrics Report

Experiment: E1_lightgbm | **Split:** Test | **Threshold (frozen):** 0.616521 **Status:** Additive extension — E1 model and predictions are IMMUTABLE

1. Bootstrap Confidence Intervals (Test Set)

Targeting 2,000 valid bootstrap samples. n_valid reported per metric.

Metric	Point Estimate	95% CI Lower	95% CI Upper	N Valid
PR-AUC	0.5315	0.5126	0.5504	2000
ROC-AUC	0.8989	0.8924	0.9055	2000
F1	0.5337	0.5179	0.5496	2000
Recall	0.4934	0.4751	0.5120	2000
Precision	0.5813	0.5624	0.6006	2000

Bootstrap samples with zero positive-class examples are skipped and re-drawn. All future experiment comparisons (E2 GRU, E1b encoding, etc.) should use CIs for significance assessment.

2. Recall at Fixed FPR Operating Points

Definition: Recall@ α = max{ TPR(t) : empirical_FPR(t) $\leq$ α } over all ROC thresholds.

FPR Target	Recall (Test)	Recall (Validation)
0.1%	0.2112	0.2628
0.5%	0.3836	0.4274
1.0%	0.4618	0.5017
2.0%	0.5306	0.5870

A 1% FPR on the test set (~74,784 legit transactions) corresponds to ~748 incorrectly flagged legitimate transactions.

3. Precision at Top-K

Definition: Of the top-K transactions ranked by predicted fraud probability, what fraction are actually fraud?

K (investigations)	Precision (Test)	Precision (Validation)	Fraud in Top-K (est.)
100	0.9800	0.9700	~98

K (investigations)	Precision (Test)	Precision (Validation)	Fraud in Top-K (est.)
500	0.9080	0.9360	~454
1,000	0.8410	0.8640	~841
5,000	0.3450	0.3594	~1725

Useful for fraud teams with limited daily investigation capacity. Compare against baseline (random) precision = fraud_rate $\approx$ 0.035.

4. Calibration Analysis

Metric	Test	Validation	Naive Baseline
Brier Score	0.0323	0.0285	0.0341
ECE (10 bins)	0.0524	0.0515	N/A

Brier Score: Lower is better. Naive baseline = fraud_rate $\times$ (1 – fraud_rate). **ECE:** Expected Calibration Error — weighted mean gap between predicted probability and actual positive fraction per bin. Perfect = 0.

See [experiments/E1_lightgbm/calibration_curve_test.png](#) for the reliability diagram.